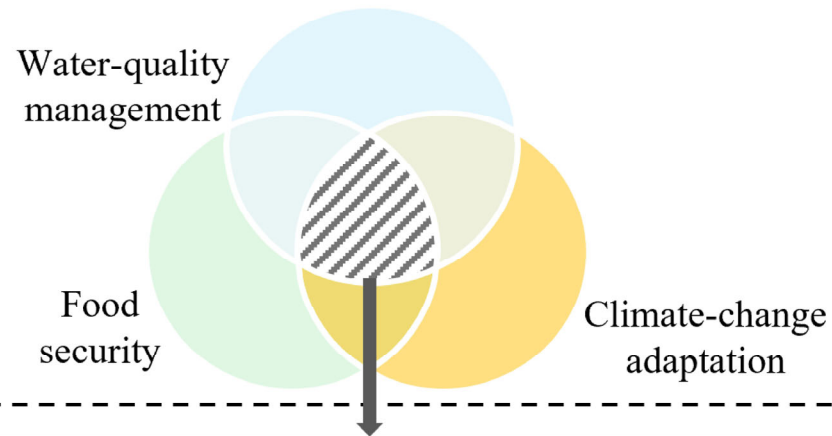


Research interests



Over-arching research goal: Probabilistic cross-scale nutrient assessment and management

Short-term goals

Long-term goals

Waterbodies

- Estimate Internal nutrient load
- Assess eutrophication under changing climate
- Co-develop adaptive strategies

Watersheds

- Characterize watershed C-N-P loads
- Project C-N-P export trajectories
- Identify intervention points

Regional

- Develop state-wide nutrient budgets
- Construct diverse eutrophication models
- Identify state-wide critical zones

National

- Develop national nutrient budgets
- Construct ensemble eutrophication-related projections
- Evaluate nation-wide management scenarios and policy pathways

Methodological foundation



Field monitoring



Hybrid modeling



Geospatial analysis



Uncertainty quantification

Future research plan

Academic & industrial collaboration

Stakeholder engagement

Knowledge transfer